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Front Bumper Assembly Support in Vehicles

Abstract

A vehicle that includes: a vehicle body; a reinforcement member that is supported by the vehicle body and which is configured to absorb forces during an impact with the vehicle; a front bumper assembly that is supported by the vehicle body and which includes front lamps; an apron bracket that is supported by the vehicle body and which extends into the front bumper assembly; and support brackets that extends between the reinforcement member and the apron bracket to thereby inhibit movement of the front bumper assembly in relation to the vehicle body.

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Background/Summary

TECHNICAL FIELD

[0001] The present application relates to supporting the front bumper assembly of a vehicle in order to inhibit (if not entirely prevent) relative movement between the front bumper assembly and the body of the vehicle.

BACKGROUND

[0002] The front bumper assembly of a vehicle includes a variety of components such as, for example, a fascia subassembly, a grille subassembly, and front lamps (e.g., headlights). Depending on the configuration of the vehicle, the weight of the front bumper assembly can be substantial, creating the potential for movement of the front bumper assembly in relation to the vehicle body during travel, which can result in damage (e.g., cracking of the grille subassembly).

[0003] In order to inhibit (if not entirely prevent) any such damage, the present disclosure describes various structures and methods for increasing support of the front bumper assembly.

SUMMARY

[0004] In one aspect of the present disclosure, a vehicle is disclosed that includes: a vehicle body; a reinforcement member; a front bumper assembly; an apron bracket; and support brackets.

[0005] The reinforcement member is supported by the vehicle body and is configured to absorb forces during an impact with the vehicle.

[0006] The front bumper assembly is supported by the vehicle body and includes: a fascia subassembly; front lamps that are supported by the fascia subassembly; and a grille subassembly that is supported by the fascia subassembly.

[0007] The apron bracket is supported by the vehicle body and the fascia subassembly such that the apron bracket extends therebetween to thereby strengthen the fascia subassembly and improve pedestrian impact performance.

[0008] The support brackets extend between the reinforcement member and the apron bracket to thereby inhibit movement of the front bumper assembly in relation to the vehicle body.

[0009] In certain embodiments, the support brackets may include generally S-shaped configurations.

[0010] In certain embodiments, the support brackets may be generally identical in configuration.

[0011] In certain embodiments, the support brackets may include: trunks; first flanges that extend from the trunks in generally orthogonal relation thereto; and second flanges that extend from the trunks in generally orthogonal relation thereto.

[0012] In certain embodiments, the first flanges may extend in a first direction, and the second flanges may extend in a second direction that is opposite to the first direction.

[0013] In certain embodiments, the first flanges may include retainers that are configured to receive first mechanical fasteners to thereby connect the support brackets to the apron bracket.

[0014] In certain embodiments, the retainers may be fixedly connected to the first flanges.

[0015] In certain embodiments, the second flanges may include apertures that extend therethrough and which are configured to receive second mechanical fasteners to thereby connect the support brackets to the reinforcement member.

[0016] In certain embodiments, the second flanges may further include projections that extend outwardly therefrom and which are configured for engagement with the reinforcement member to inhibit rotation of the support brackets in relation to the reinforcement member and the apron bracket.

[0017] In certain embodiments, the reinforcement member may include openings that are configured to receive the projections.

[0018] In another aspect of the present disclosure, a vehicle is disclosed that includes: a vehicle body; a reinforcement member that is supported by the vehicle body and which is configured to absorb forces during an impact with the vehicle; a front bumper assembly that is supported by the vehicle body and which includes front lamps; an apron bracket that is supported by the vehicle

body and which extends into the front bumper assembly; and support brackets that extend between the reinforcement member and the apron bracket to thereby inhibit movement of the front bumper assembly in relation to the vehicle body.

[0019] In certain embodiments, the support brackets may include outer support brackets that are positioned in general alignment with the front lamps along a width of the vehicle.

[0020] In certain embodiments, the support brackets may further include at least one inner support bracket that is positioned between the outer support brackets.

[0021] In certain embodiments, the at least one inner support bracket may be generally centered along the width of the vehicle.

[0022] In certain embodiments, the support brackets may include trunks and upper and lower flanges that extend from the trunks in generally opposite directions and in generally parallel relation.

[0023] In another aspect of the present disclosure, a method of assembling a vehicle is disclosed. The method includes: connecting an apron bracket to a vehicle body; connecting a reinforcement member to the vehicle body, wherein the reinforcement member is configured to absorb impact forces; connecting support brackets to the apron bracket and the reinforcement member such that the support brackets extend therebetween; and connecting a front bumper assembly to the vehicle body such that the apron bracket extends into the front bumper assembly, whereby the support brackets inhibit movement of the front bumper assembly in relation to the vehicle body.

[0024] In certain embodiments, connecting the support brackets to the apron bracket and the reinforcement member may include connecting first flanges to the apron bracket and connecting second flanges to the reinforcement member such that the first flanges and the second flanges extend in generally opposite directions.

[0025] In certain embodiments, connecting the support brackets to the reinforcement member may include inserting projections that extend outwardly from the support brackets into openings in the reinforcement member to thereby inhibit rotation of the support brackets in relation to the reinforcement member and the apron bracket.

[0026] In certain embodiments, connecting the support brackets to the apron bracket and the reinforcement member may include connecting outer support brackets to the apron bracket and the reinforcement member such that the outer support brackets are generally aligned with front lamps of the vehicle.

[0027] In certain embodiments, connecting the support brackets to the reinforcement member and the apron bracket may include connecting at least one inner support bracket to the reinforcement member and the apron bracket such that the at least one inner support bracket is positioned between the outer support brackets.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] According to common practice, the various features of the drawings may not be to scale and may be arbitrarily expanded or reduced for clarity.

[0029] FIG. 1 is a partial, top, perspective view of a vehicle according to the principles of the present disclosure.

[0030] FIG. 2 is a partial, front, perspective view of the vehicle illustrating a bumper beam, an apron bracket, and support brackets.

[0031] FIG. 3 is a partial, side, plan view of the vehicle illustrating the bumper beam, the apron bracket, and the support brackets.

[0032] FIG. 4 is a front, perspective view of one of the support brackets.

DETAILED DESCRIPTION

[0033] The present disclosure describes a vehicle that includes: a vehicle body; a reinforcement member, which is configured to absorb forces during an impact with the vehicle; a front bumper assembly; an apron bracket, which extends into the front bumper assembly; and support brackets, which extend between the reinforcement member and the apron bracket. The support brackets inhibit (if not entirely prevent) relative movement between the front bumper assembly and the body of the vehicle during travel and, thus, damage to the front bumper assembly that may otherwise occur.

[0034] With reference to FIGS. 1-4, a vehicle **10** is disclosed that includes: a vehicle body **100**; a reinforcement member **200** (e.g., a crash beam **202**); a front bumper assembly **300**, which is positioned (located) at a front-end **12** of the vehicle **10**; an apron bracket **400**; and a plurality of support brackets (stays) **500**. Although generally illustrated and described in the context of a passenger vehicle, it should be appreciated that the various components described herein may be configured for use with a wide variety of vehicles (e.g., trucks, SUVs, vans, buses, etc.).

[0035] The vehicle body **100** is a structural component that not only provides strength and rigidity to the vehicle **10** but supports and provides a framework for the reinforcement member **200**, the front bumper assembly **300**, the apron bracket **400**, and various other components.

[0036] The reinforcement member **200** protects and fortifies the front-end **12** of the vehicle **10** and is positioned (located) rearwardly of the front bumper assembly **300**. The reinforcement member **200** is connected (secured, mounted) to (or otherwise supported by) the vehicle body **100** and is configured to absorb force during an impact with the vehicle **10** (e.g., a crash, a collision, etc.). It is envisioned that the reinforcement member **200** may be removably or fixedly (i.e., non-removably) connected to the vehicle body **100** in any suitable manner, including, for example, via mechanical fasteners (e.g., bolts, screws, rivets, pins, clips, or the like), via welding, via integral (e.g., unitary, monolithic) formation therewith, etc.

[0037] The front bumper assembly **300** is connected (secured, mounted) to (or otherwise supported by) the vehicle body **100** and extends laterally (e.g., horizontally) along a width **W** (FIG. 1) of the vehicle **10** (i.e., across the front-end **12**). The front bumper assembly **300** includes: a fascia subassembly **302**; front lamps **304** (e.g., headlights **306**), which are connected (secured, mounted) to (or otherwise supported by) the fascia subassembly **302**; and a grille subassembly **308**.

[0038] The fascia subassembly **302** is positioned (located) and extends axially forward of the vehicle body **100** and the reinforcement member **200**. The fascia subassembly **302** supports the front lamps **304** and serves a variety of functions including, for example, improving aerodynamics, contributing to the overall aesthetic appearance of the vehicle **10**, and concealing (covering) various components of the vehicle **10**.

[0039] The grille subassembly **308** is connected (secured, mounted) to (or otherwise supported by) the fascia subassembly **302**. Like the fascia subassembly **302**, the grille subassembly **308** serves a variety of functions including, for example, contributing to the overall aesthetic appearance of the vehicle **10** and facilitating airflow into the engine compartment.

[0040] The apron bracket **400** is connected (secured, mounted) to (or otherwise supported by) the vehicle body **100** and the front bumper assembly **300** (i.e., the fascia subassembly **302**) and extends therebetween. More specifically, the apron bracket **400** extends into the front bumper assembly **300** (i.e., the fascia subassembly **302**), whereby the apron bracket **400** strengthens the fascia subassembly **302** to improve pedestrian impact performance.

[0041] The support brackets **500** extend between the reinforcement member **200** and the apron bracket **400** and are fixedly (i.e., non-movably) connected (secured, mounted) thereto, as described in further detail below. The support brackets **500** inhibit (if not entirely prevent) movement of the front bumper assembly **300** in relation to the vehicle body **100** and, thus, damage to the front bumper assembly **300** (e.g., cracking of the grille subassembly **308**) that may otherwise occur during travel of the vehicle **10**.

[0042] With reference to FIGS. 2-4 in particular, the support brackets **500** will be discussed. The

vehicle **10** includes outer support brackets **500O** and (one or more) at least one inner support bracket **500I**. Although shown as including three support brackets **500** in the illustrated embodiment (i.e., a pair of outer support brackets **500Oi**, **500Oii** and a single inner support bracket **500I**), it is envisioned that the particular number of support brackets **500** may be increased or decreased without departing from the scope of the present disclosure. For example, embodiments in which the vehicle **10** may include two or more inner support brackets **500I** also envisioned herein, as are embodiments in which the vehicle **10** may be devoid of the inner support brackets **500I** altogether (i.e., embodiments that exclusively include the outer support brackets **500O**).

[0043] The outer support brackets **500Oi**, **500Oii** are positioned in general alignment with the front lamps **304** (FIG. **1**) (i.e., along the width **W** of the vehicle **10**), which allows the outer support brackets **500O** to support the weight thereof and thereby further inhibit (if not entirely prevent) movement of the front bumper assembly **300** in relation to the vehicle body **100**.

[0044] As seen in FIG. **2**, the inner support bracket **500I** is positioned (located) between the outer support brackets **500Oi**, **500Oii**. More specifically, in the illustrated embodiment, the inner support bracket **500I** is generally centered along the width **W** (FIG. **1**) of the vehicle **10**. Embodiments in which the inner support bracket **500I** may be eccentrically positioned (i.e., such that the inner support bracket **500I** is located off-center along the width **W** of the vehicle **10**) are also envisioned herein (e.g., to create clearance for additional components of the vehicle **10**), however, and would not be beyond the scope of the present disclosure.

[0045] The support brackets **500** are generally identical in configuration and include trunks **502** (FIG. **4**) and respective lower (first) and upper (second) flanges **504**, **506**, which extend from the trunks **502** in generally opposite directions, as described in further detail below. More specifically, the support brackets **500** are generally S-shaped in configuration, which creates increased clearance (e.g., vis-à-vis a generally C-shaped configuration) for tooling during installation of the support brackets **500** and connection to the reinforcement member **200** and the apron bracket **400**.

[0046] The trunks **502** extend between and connect the respective lower and upper flanges **504**, **506** and are generally linear in configuration. More specifically, the trunks **502** include lower (first) ends **508**, which are connected (secured) to the lower flanges **504**, and upper (second) ends **510**, which are connected (secured) to the upper flanges **506**.

[0047] In the illustrated embodiment, the trunks **502** include one or more strengthening members **512** (FIG. **4**) (e.g., beads **514**, ribs, etc.) and indicators **516** (e.g., arrows **518**), which identify the direction of installation. Embodiments of the support brackets **500** that are devoid of the strengthening member(s) **512** and/or the indicators **516** are also envisioned herein, however, and would not be beyond the scope of the present disclosure.

[0048] The lower flanges **504** extend from the trunks **502** in generally orthogonal (perpendicular) relation thereto and in a first (i.e., forward) direction. The lower flanges **504** include retainers **520** (FIG. **4**) that are configured to receive (first) mechanical fasteners **522** (FIG. **3**) (e.g., screws, bolts, etc.) such that the mechanical fasteners **522** extend through the support brackets **500** (via the retainers **520**) and into apertures **402** in the apron bracket **400** to thereby connect the support brackets **500** to the apron bracket **400**. More specifically, in the illustrated embodiment, the retainers **520** are configured as weld nuts **524** that are fixedly (i.e., non-movably and non-removably) connected to the lower flanges **504**.

[0049] The upper flanges **506** extend from the trunks **502** in generally orthogonal (perpendicular) relation thereto and in a second (i.e., rearward) direction, which is generally opposite to the first direction. More specifically, the support brackets **500** are configured such that the respective lower and upper flanges **504**, **506** extend from the trunks **502** in generally parallel relation.

[0050] The upper flanges **506** include apertures **526** and projections **528** (FIGS. **3**, **4**) (e.g., tabs **530**).

[0051] The apertures **526** extend through the upper flanges **506** and are configured to receive (second) mechanical fasteners **532** (FIG. **3**) (e.g., screws, bolts, etc.) such that the mechanical

fasteners **532** extend through the support brackets **500** (via the apertures **526**) and into apertures **204** in the reinforcement member **200** to thereby connect the support brackets **500** to the reinforcement member **200**.

[0052] The projections **528** (FIG. **4**) are generally U-shaped in configuration and extend outwardly (i.e., rearwardly and vertically) from the upper flanges **506**. The projections **528** are configured for engagement (contact) with the reinforcement member **200** in order to inhibit rotation of the support brackets **500** in relation to the reinforcement member **200** and the apron bracket **400** upon the application of torque thereto (e.g., during connection of the mechanical fasteners **522**). More specifically, the projections **528** are configured for insertion into corresponding openings **206** (FIG. **3**) (e.g., slots **208**) that are formed in (defined by) the reinforcement member **200**, which facilitates installation of the support brackets **500** by maintaining alignment between the retainers **520** (FIG. **4**) and the apertures **402** (FIG. **3**) in the apron bracket **400**.

[0053] With continued reference to FIGS. **1-4**, a method of assembling the vehicle **10** will be discussed.

[0054] Initially, the reinforcement member **200** (FIG. **1**) and the apron bracket **400** are connected to the vehicle body **100**. Thereafter, the support brackets **500** are connected to the apron bracket **400** and the reinforcement member **200** such that the outer support brackets **500O** (FIG. **2**) are generally aligned with the front lamps **304** (FIG. **1**). More specifically, during connection of the support brackets **500**, the projections **528** (FIGS. **3, 4**) are inserted into the openings **206** (FIG. **3**) in the reinforcement member **200**, which aligns the retainers **520** (FIG. **4**) and the apertures **526** with the apertures **402, 204** (FIG. **3**) in the apron bracket **400** and the reinforcement member **200**, respectively, and inhibits rotation of the support brackets **500** in relation to the apron bracket **400** and the reinforcement member **200**, thereby maintaining proper alignment. The mechanical fasteners **522** (FIG. **3**) are then inserted through the retainers **520** and into the apertures **402** in the apron bracket **400** in order to connect the lower flanges **504** thereto, and the mechanical fasteners **532** are inserted through the apertures **526** and into the apertures **204** in the reinforcement member **200** in order to connect the upper flanges **506** thereto. As indicated above, upon connection of the support brackets **500** to the apron bracket **400** and the reinforcement member **200**, the flanges **504, 506** extend in generally opposite directions and in generally parallel relation to each other.

[0055] Either prior or subsequent to connection of the support brackets **500**, the front bumper assembly **300** (FIG. **1**) is connected to the vehicle body **100** such that the apron bracket **400** extends into the fascia subassembly **302**.

[0056] Persons skilled in the art will understand that the various embodiments of the disclosure described herein and shown in the accompanying figures constitute non-limiting examples, and that additional components and features may be added to any of the embodiments discussed herein above without departing from the scope of the present disclosure. Additionally, persons skilled in the art will understand that the elements and features shown or described in connection with one embodiment may be combined with those of another embodiment without departing from the scope of the present disclosure and will appreciate further features and advantages of the presently disclosed subject matter based on the description provided. Variations, combinations, and/or modifications to any of the embodiments and/or features of the embodiments described herein that are within the abilities of a person having ordinary skill in the art are also within the scope of the disclosure, as are alternative embodiments that may result from combining, integrating, and/or omitting features from any of the disclosed embodiments.

[0057] Use of broader terms such as “comprises,” “includes,” and “having” should be understood to provide support for narrower terms such as “consisting of,” “consisting essentially of,” and “comprised substantially of.” Accordingly, the scope of protection is not limited by the description set out above but is defined by the claims that follow and includes all equivalents of the subject matter of the claims.

[0058] In the preceding description, reference may be made to the spatial relationship between the

various structures illustrated in the accompanying drawings, and to the spatial orientation of the structures. However, as will be recognized by those skilled in the art after a complete reading of this disclosure, the structures described herein may be positioned and oriented in any manner suitable for their intended purpose. Thus, the use of terms such as “above,” “below,” “upper,” “lower,” “inner,” “outer,” “left,” “right,” “upward,” “downward,” “inward,” “outward,” etc., should be understood to describe a relative relationship between the structures and/or a spatial orientation of the structures. Those skilled in the art will also recognize that the use of such terms may be provided in the context of the illustrations provided by the corresponding figure(s).

[0059] Additionally, terms such as “approximately,” “generally,” “substantially,” and the like should be understood to allow for variations in any numerical range or concept with which they are associated and encompass variations on the order of 25% (e.g., to allow for manufacturing tolerances and/or deviations in design). For example, the term “generally parallel” should be understood as referring to configurations in which the pertinent components are oriented so as to define an angle therebetween that is equal to $180^{\circ} \pm 25\%$ (i.e., an angle that lies within the range of (approximately) 135° to (approximately) 225°) and the term “generally orthogonal” should be understood as referring to configurations in which the pertinent components are oriented so as to define an angle therebetween that is equal to $90^{\circ} \pm 25\%$ (i.e., an angle that lies within the range of (approximately) 67.5° to (approximately) 112.5°). The term “generally parallel” should thus be understood as referring to encompass configurations in which the pertinent components are arranged in parallel relation, and the term “generally orthogonal” should thus be understood as referring to encompass configurations in which the pertinent components are arranged in orthogonal relation.

[0060] Although terms such as “first,” “second,” “third,” etc., may be used herein to describe various operations, elements, components, regions, and/or sections, these operations, elements, components, regions, and/or sections should not be limited by the use of these terms in that these terms are used to distinguish one operation, element, component, region, or section from another. Thus, unless expressly stated otherwise, a first operation, element, component, region, or section could be termed a second operation, element, component, region, or section without departing from the scope of the present disclosure.

[0061] Each and every claim is incorporated as further disclosure into the specification and represents embodiments of the present disclosure. Also, the phrases “at least one of A, B, and C” and “A and/or B and/or C” should each be interpreted to include only A, only B, only C, or any combination of A, B, and C.

Claims

1. A vehicle comprising: a vehicle body; a reinforcement member supported by the vehicle body and configured to absorb forces during an impact with the vehicle; a front bumper assembly supported by the vehicle body, wherein the front bumper assembly includes: a fascia subassembly; front lamps supported by the fascia subassembly; and a grille subassembly supported by the fascia subassembly; an apron bracket supported by the vehicle body and the fascia subassembly such that the apron bracket extends therebetween to thereby strengthen the fascia subassembly and improve pedestrian impact performance; and support brackets extending between the reinforcement member and the apron bracket to thereby inhibit movement of the front bumper assembly in relation to the vehicle body.
2. The vehicle of claim 1, wherein the support brackets include generally S-shaped configurations.
3. The vehicle of claim 2, wherein the support brackets are generally identical in configuration.
4. The vehicle of claim 2, wherein the support brackets include: trunks; first flanges extending from the trunks in generally orthogonal relation thereto; and second flanges extending from the trunks in generally orthogonal relation thereto.

5. The vehicle of claim 4, wherein the first flanges extend in a first direction, and the second flanges extend in a second direction opposite to the first direction.
6. The vehicle of claim 4, wherein the first flanges include retainers configured to receive first mechanical fasteners to thereby connect the support brackets to the apron bracket.
7. The vehicle of claim 6, wherein the retainers are fixedly connected to the first flanges.
8. The vehicle of claim 6, wherein the second flanges include apertures extending therethrough configured to receive second mechanical fasteners to thereby connect the support brackets to the reinforcement member.
9. The vehicle of claim 8, wherein the second flanges further include: projections extending outwardly therefrom and configured for engagement with the reinforcement member to inhibit rotation of the support brackets in relation to the reinforcement member and the apron bracket.
10. The vehicle of claim 9, wherein the reinforcement member includes openings configured to receive the projections.
11. A vehicle comprising: a vehicle body; a reinforcement member supported by the vehicle body and configured to absorb forces during an impact with the vehicle; a front bumper assembly supported by the vehicle body and including front lamps; an apron bracket supported by the vehicle body and extending into the front bumper assembly; and support brackets extending between the reinforcement member and the apron bracket to thereby inhibit movement of the front bumper assembly in relation to the vehicle body.
12. The vehicle of claim 11, wherein the support brackets include outer support brackets positioned in general alignment with the front lamps along a width of the vehicle.
13. The vehicle of claim 12, wherein the support brackets further include: at least one inner support bracket positioned between the outer support brackets.
14. The vehicle of claim 13, wherein the at least one inner support bracket is generally centered along the width of the vehicle.
15. The vehicle of claim 11, wherein the support brackets include: trunks; and upper and lower flanges extending from the trunks in generally opposite directions and in generally parallel relation.
16. A method of assembling a vehicle, the method comprising: connecting an apron bracket to a vehicle body; connecting a reinforcement member to the vehicle body, wherein the reinforcement member is configured to absorb impact forces; connecting support brackets to the apron bracket and the reinforcement member such that the support brackets extend therebetween; and connecting a front bumper assembly to the vehicle body such that the apron bracket extends into the front bumper assembly, whereby the support brackets inhibit movement of the front bumper assembly in relation to the vehicle body.
17. The method of claim 16, wherein connecting the support brackets to the apron bracket and the reinforcement member includes connecting first flanges to the apron bracket and connecting second flanges to the reinforcement member such that the first flanges and the second flanges extend in generally opposite directions.
18. The method of claim 16, wherein connecting the support brackets to the reinforcement member includes inserting projections extending outwardly from the support brackets into openings in the reinforcement member to thereby inhibit rotation of the support brackets in relation to the reinforcement member and the apron bracket.
19. The method of claim 16, wherein connecting the support brackets to the apron bracket and the reinforcement member includes connecting outer support brackets to the apron bracket and the reinforcement member such that the outer support brackets are generally aligned with front lamps of the vehicle.
20. The method of claim 19, wherein connecting the support brackets to the reinforcement member and the apron bracket includes connecting at least one inner support bracket to the reinforcement member and the apron bracket such that the at least one inner support bracket is positioned between the outer support brackets.

